Reg. No. | | | | | | | | | | | [18ECDE545]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

INTERNET OF THINGS

Time:3 Hours **Maximum Marks:70**

Instructions: i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.

ii. Missing data may be suitably assumed.

1.	**a.**	What is IoT? Explain components of IoT with a suitable example	**04**
	b.	**i.** Explain the role of internet and digitisation in the development of IoT	**05**
		ii. highlight the features of IoT based connected roadways	**05**
2.	**a.**	Illustrate the interaction of sensors and actuators with the physical world with a suitable example	**7**
	b.	Demonstrate various IoT access technologies with respect to range of communication.	**07**
3.	**a.**	What is IoT data analytics and describe the challenges?	**07**
	b.	Describe 4 different types of data analytics and sketch the graph rating the complexity	**07**

OR

4.	**a.**	Demonstrate the common application domains of Machine learning for IoT with examples?	**07**
	b.	Write short notes on Hadoop for IoT	**07**
5.	**a.**	Explain the specifications and functional blocks of Arduino Uno in detail.	**07**
	b.	Explain any five Arduino function libraries.	**07**

OR

6.	**a.**	With a schematic of interfacing LED to Arduino board, write a sketch (code) to blink an LED light continuously connected to pin number 12 of the Arduino Uno	**07**
	b.	With a schematic of interfacing DHT to Arduino board, Write a sketch to read the temperature values	**07**
7.	**a.**	With a schematic diagram of Raspberry pi board, explain functional blocks and specifications.	**07**
	b.	Illustrate the loading of operating system and initial configurations of Raspberry pi that are necessary to perform when using a new raspberry pi board.	**07**

OR

8.	**a.**	Write a python program to interface an LED to Raspberry pi and blink LED thrice with duration of 3 seconds each.	**08**
	b.	Suggest two of methods of capturing image using pi camera interfaced to Raspberry pi	**06**